
Chapter 6: Crease Stabilization and Early Stopping

(Experiment 3)

A Technical Report from the Puno Calculus Research Program

Michael Grafiel Sayson Puno | July 2026

Question

Can crease density stabilization serve as an early stopping criterion? When units stop toggling between on/off states, has the network's partition converged?

The Puno Calculus predicts that when crease density stabilizes (the rate of change drops near zero), the folding pattern has converged and further training is unnecessary.

Setup

- **Problem:** Multi-scale checkerboard (60/20/20 split)
- **Three architectures:** 3-layer, 4-layer, 6-layer MLPs
- **Three early stopping strategies:**
 - **Full:** Train 500 epochs (baseline)
 - **Val loss plateau:** Stop after 25 epochs without val loss improvement
 - **Crease stabilization:** Stop after 15 epochs where $|\Delta \text{crease}| < 1.5\%$ of current value
 - **Both:** Stop when either triggers

Results

Architecture	Crease Stop	Val Stop	Full Training
Shallow (3L)	ep 101, 78.1%	ep 369, 87.1%	ep 500, 90.3%
Medium (4L)	ep 180, 88.9%	ep 265, 89.3%	ep 500, 91.5%
Deep (6L)	ep 145, 88.4%	ep 240, 90.4%	ep 500, 91.2%

Key Findings

1. **Crease density never truly plateaus** on this problem. It declines monotonically throughout training in two phases:

- **Fast phase** (~epochs 1-150): Units quickly settle into on/off states. Crease density drops rapidly.
- **Slow phase** (~epochs 150-500): The partition continues to refine gradually. Crease density declines more slowly but never stops.

2. **Crease stabilization triggers too early.** It fires at the fast-to-slow transition point, when the network has done the coarse partitioning but not the fine refinement. Stopping here loses 10+ points of accuracy.

3. **Val loss plateau also triggers prematurely** — it loses 3-6% test accuracy vs. full training. The shallow network is hit hardest: val loss stops at ep 369 with 87.1% vs full 90.3%.

4. **Crease density is a more sensitive measure of ongoing learning than validation metrics.** It continues to decline long after val loss has plateaued. The rate of crease density decline correlates with continued accuracy gains.

Interpretation

Early stopping via crease stabilization does not work as a standalone binary trigger on this problem. The network's partition continues to refine throughout training, and crease density captures this ongoing refinement that val loss misses.

However, this negative result reveals a positive finding: **crease density slope is a label-free proxy for whether the network is still learning.** This is useful in settings where labels are unavailable (self-supervised learning, online learning) or where validation metrics are noisy.

A principled early stopping rule would combine both signals: stop when BOTH val loss is stable AND crease density decline rate is near zero. This would prevent the premature stopping that val-loss-only rules produce on this problem.
